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CrisisAid Route Planner Report

Description of the Project

The objective of this project is to design and implement a web-based application that supports decision-making in humanitarian crisis scenarios. The system is called CrisisAid Route Planner, which allows users to select a starting location and a destination and gives the most efficient route between them. The application models real-world challenges such as limited resources, unsafe roads, and time pressure by assigning costs to routes. The goal is to demonstrate a simple way on how algorithmic solutions can be applied to optimize routing decisions under constraints.

Significance of the Project

This project is important because it shows how algorithms can help make decisions in difficult situations like wars or disasters. In real life, people need to find safe and efficient routes to deliver supplies, but conditions are uncertain and time is limited. Although this program uses simple labels like A, B, C, and D instead of real locations, it represents a real-world system in a simplified way. The numbers used in the program represent cost, which can stand for distance, danger, or travel difficulty. This simplified model still demonstrates how an algorithm can compare multiple routes and choose the best one. It shows how decision-making can be improved using computational methods instead of guessing.

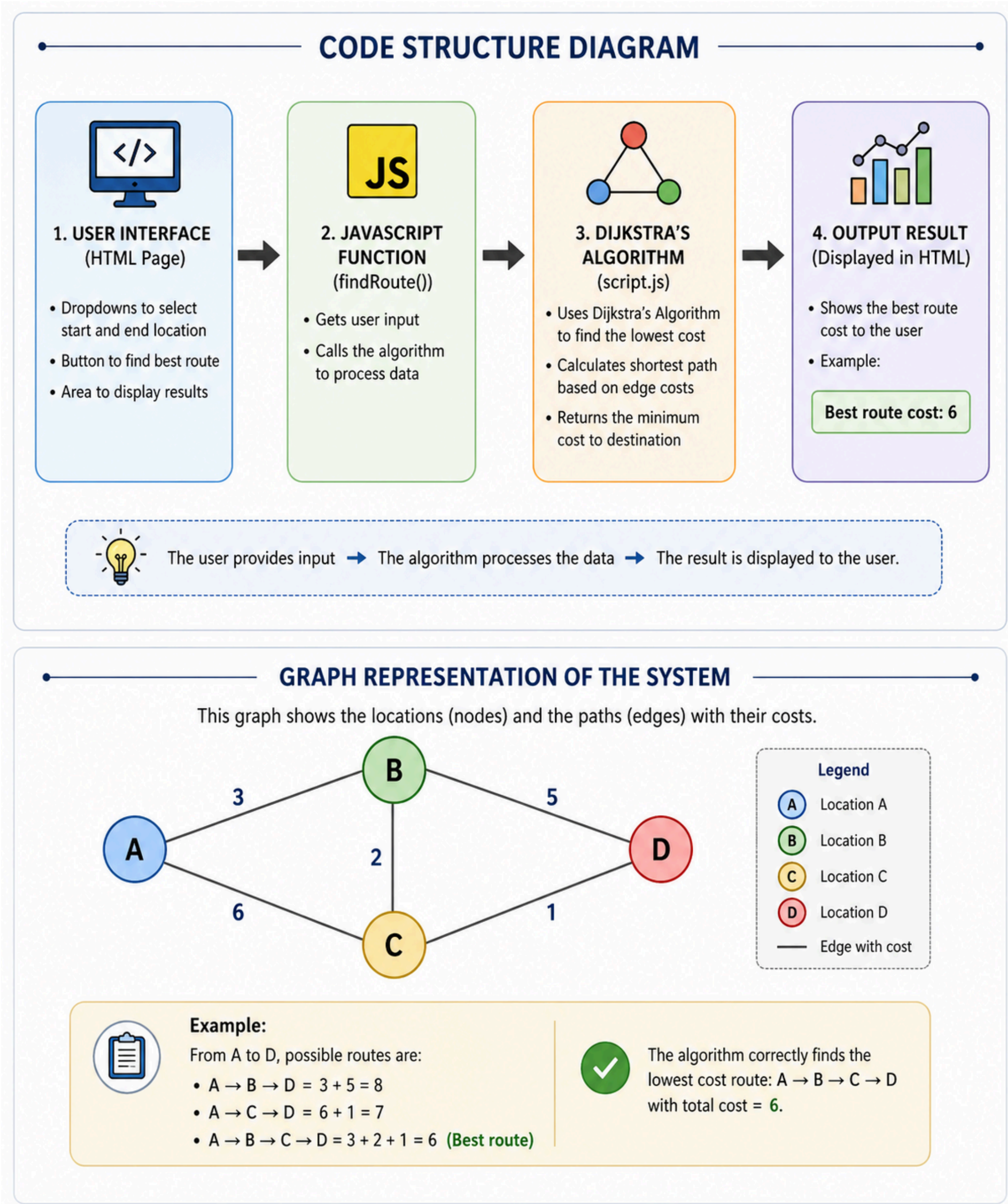
Code Structure

The project follows a simple and organized structure with three main components:

- **index.html**: Handles the structure of the application, including user inputs and buttons.
- **style.css**: Provides styling and layout to ensure a clean and user-friendly interface.
- **app.js**: Contains the algorithm implementation and logic for processing user input and displaying results.

Code Structure Diagram

This structure explains how the diagram works, making the system easy to understand and maintain.



Description of Algorithms

The project uses Dijkstra's Algorithm, a well-known algorithm for finding the shortest path between nodes in a graph. In this system nodes represent locations (A, B, C, D), edges represent roads connecting locations, each edge has a cost that represents distance and risk combined. The algorithm begins at the starting node and assigns an initial cost of zero. It then repeatedly selects the unvisited node with the smallest cost and updates the costs of its neighboring nodes. This process continues until the destination node is reached. The final result is the minimum-cost path between the selected locations.

Verification of Algorithms

To verify the correctness of the algorithm, a test case was used:

- A → B = 3
- A → C = 6
- B → C = 2
- B → D = 5
- C → D = 1

Possible routes from A to D:

- A → C → D = 7
- A → B → D = 8
- A → B → C → D = 6

The algorithm correctly selects the path A → B → C → D with a cost of 6, which is the minimum among all options. This confirms that the implementation of Dijkstra's Algorithm is functioning correctly.

Execution Results and Analysis

The application was tested with multiple inputs, and it consistently produced correct results. When users select a starting and ending location, the system calculates and displays the optimal route cost.

For example:

-Input: A → D

-Output: Best route cost = 6

This demonstrates that the system can effectively evaluate multiple possible paths and select the most efficient one. The results highlight the usefulness of algorithmic decision-making in situations where manual evaluation would be inefficient or error-prone. Although the current system uses a simplified dataset, it shows how more complex real-world data could be incorporated to improve decision-making in crisis response scenarios.

Conclusion

This project successfully demonstrates how algorithmic techniques can be applied to real-world humanitarian challenges. The CrisisAid Route Planner uses Dijkstra's Algorithm to determine the most efficient route under constraints such as distance and risk. Through this project, key concepts such as graph modeling, optimization, and decision-making under uncertainty were explored. One limitation of the system is that it uses a small predefined dataset and does not include real-time data or dynamic updates. Future improvements could include adding real maps, user-defined inputs, and more advanced algorithms. Overall, the project shows how computer science concepts learned in class can be applied to meaningful real-world problems.

AI Usage

AI tools were used to assist in the development of this project. Specifically, AI was used for helping in giving specific information on how this system is supposed to work. Any debugging to help with understanding certain words were also used with AI. AI was used as a support tool to make sure that any unknown information was clearly explained. Also, AI helped in the development of the diagram which was made to show the structure of the program.

GitHub

All project components, including the HTML, CSS, JavaScript files, and documentation, have been uploaded to a public repository on GitHub. The repository is accessible via the submitted link and demonstrates proper organization of project files. This ensures that the project is easy to access, review, and run.

Overall Quality of Report

This report demonstrates a clear understanding of algorithmic design and its application to real-world problems. The project is directly related to course concepts, including graph algorithms and optimization. Each section is clearly explained, and the structure follows the required guidelines. The writing is organized, and accurate, reflecting careful effort in both implementation and documentation.